

KARAN JADHAV

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PROFESSIONAL SUMMARY

AI/ML Engineer with 6 months of production experience, who shipped a multi-agent LangGraph chatbot now handling procurement automation at Nova Techset. Specialist in Generative AI, Agentic AI systems, RAG pipelines, LLM fine-tuning, and scalable backend APIs. Proven track record building scalable solutions using LangGraph, LangChain, FastAPI, PyTorch, and MLOps tooling. Delivered 92% CV accuracy and NDCG@5 of 0.93 on live vendor ranking systems.

PROFESSIONAL EXPERIENCE

AI/ML Intern – Nova Techset Limited, Pune

Nov 2025 – May 2026

Full Time

Project: P2P Procure-to-Pay AI Chatbot — Multi-Agent System

- Architected **multi-agent LangGraph chatbot** automating **80%** of procurement workflows across PR management, product search, and Q&A queries.
- Designed **orchestrator routing across 5 agents**, cutting misrouted queries by **40%** and response latency by **1.2s**
- Developed **Product Search Agent** with context-aware follow-ups and **LLM-Q&A** for product and platform queries
- Built and deployed **FastAPI REST APIs** for model serving and system integration.
- Developed **Vendor Recommendation Engine** using **LambdaMART, XGBoost, CatBoost** with **NDCG optimization**; achieved **NDCG@5 of 0.9364 (train) / 0.9336 (val)** with minimal overfitting.
- Monitored pipelines using **Grafana** for observability and performance tracking

Other Contributions:

- Built **OCR + LLM pipeline** processing at **85%+** extraction accuracy, saving **-12 hrs/week** of manual work
- Collaborated with **frontend, DevOps, QA** and resolved production issues across lifecycle

Tech Stack: *Python, LangGraph, LangChain, FastAPI, Streamlit, XGBoost, LightGBM (LambdaMART), LLMs, RAG, Hugging Face, OCR, MLflow, LangSmith, Docker, Git, PostgreSQL, Grafana*

TECHNICAL SKILLS

- Programming Languages:** Python, SQL, C++, OOPS
- Generative AI & LLMs:** Agentic AI, LangGraph, LangChain, Hugging Face Transformers, LLMs, RAG, VectorDB
- Machine Learning & AI:** Machine Learning, Deep Learning, NLP, Predictive Analytics, Prompt Engineering
- Tools & Frameworks:** Scikit-learn, TensorFlow, Keras, PyTorch, NumPy, Pandas, Flask, FastAPI, Docker, Git
- Data Visualization & Analytics:** Power BI, Tableau, Matplotlib, Seaborn, MS Excel, Streamlit
- Data Engineering & Cloud:** ETL, AWS, MongoDB, MySQL, PostgreSQL, Pinecone, ChromaDB, FAISS

PROJECTS

Object Classification System | Deep Learning | Python, TensorFlow, Keras, Flask

Mar 2025

- Built end-to-end **Computer Vision pipeline** using **CNN (TensorFlow/Keras)** for multi-class image classification across **47 mammal species**, achieving **92% training** and **85% validation accuracy**
- Engineered **Flask REST API** with **OpenCV** feature extraction enabling **real-time inference** and **similarity search**
- Deployed interactive web interface with drag-and-drop upload and **real-time inference visualization**

Diabetes Classification System | Machine Learning | Python, Scikit-learn, Flask

Oct 2024

- Built **ensemble ML model** (Logistic Regression, SVC, Random Forest) achieving **94% recall** and **85% F1-score**
- Performed **EDA** using Matplotlib/Seaborn identifying **glucose & BMI** as key predictors via **feature engineering**
- Deployed **production-ready Flask REST API** for real-time diabetes risk assessment with **Scikit-learn** pipeline

EDUCATION

College name: Government College of Engineering Amravati

Dec/2021 to Jun/2025

Department: CSE (CGPA:7.54)

Maharashtra, India

CERTIFICATIONS

Data Science Master By Physics Wallah (Link)

May 2024

Data Analytics Training by Trainity (Link)

Sep 2024